

PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

**Comparative Study of Monolithic and Microservices
Architectures in Cloud-Based Systems**

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Comparative Study of Monolithic and Microservices Architectures in Cloud-Based Systems**” partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Isha Kansal**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

The rapid evolution of cloud computing has significantly influenced the design and deployment of modern software systems. Applications today must support high scalability, reliability, and real-time processing to meet increasing user demands. Traditionally, monolithic architecture has been widely used due to its simplicity and ease of implementation. However, as system complexity increases, monolithic systems face challenges such as poor scalability, limited fault isolation, and difficulty in maintenance.

Microservices architecture has emerged as a modern solution that decomposes applications into smaller, loosely coupled, and independently deployable services. This architecture improves scalability, resilience, and flexibility, making it suitable for cloud-native environments. However, it introduces challenges such as increased communication overhead, operational complexity, and data consistency issues.

This report presents a comprehensive comparative analysis of monolithic and microservices architectures based on multiple parameters, including performance, scalability, resource utilization, fault tolerance, and deployment complexity. Experimental results demonstrate that monolithic systems are efficient for small-scale applications, whereas microservices architectures provide superior performance and reliability in large-scale, distributed environments.

1. INTRODUCTION

1.1 Background

Cloud computing has revolutionized the way software systems are designed and deployed. With the rise of digital services such as online banking, e-commerce, and real-time analytics, applications must handle large volumes of data and concurrent users efficiently.

Traditional systems were built using monolithic architecture, where all components are integrated into a single application. While this approach simplifies development, it becomes inefficient as the system grows in size and complexity.

1.2 Need for Modern Architectures

Modern applications require:

- High scalability

- Continuous availability
- Rapid deployment cycles
- Fault tolerance

Monolithic systems struggle to meet these requirements due to tight coupling between components. As a result, microservices architecture has gained popularity as a scalable and flexible alternative.

1.3 Problem Statement

- Monolithic systems:
 - Difficult to scale
 - High risk of system failure
 - Slow deployment process
 - Microservices systems:
 - Complex to manage
 - Increased latency due to communication
 - Requires advanced infrastructure
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1.4 Objectives

The main objectives of this study are:

- To compare monolithic and microservices architectures
 - To evaluate system performance under different workloads
 - To analyze scalability and fault tolerance
 - To study resource utilization and operational complexity
 - To recommend the most suitable architecture for cloud-based applications
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2. METHODOLOGY

2.1 Experimental Design

A controlled experimental approach is used to ensure fair comparison. Both architectures are tested using identical workloads and system configurations.

- User load range: 1000 to 10,000 concurrent users
 - Multiple test runs for accuracy
 - Average values used for analysis
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2.2 Workload Simulation

The following real-world operations are simulated:

- User authentication
- Transaction processing
- Data retrieval
- Database updates

These tasks represent typical operations in cloud-based applications such as banking systems.

2.3 Evaluation Metrics

The performance of both architectures is evaluated using:

- **Response Time:** Measures system latency
- **Throughput:** Number of processed requests per second
- **CPU Usage:** Processing power consumed
- **Memory Usage:** RAM utilization
- **Scalability:** System behavior under increasing load

- **Fault Tolerance:** Ability to handle failures
 - **Setup Time:** Initial configuration time
 - **Operational Complexity:** Ease of maintenance and deployment
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3. SYSTEM ARCHITECTURE

3.1 Monolithic Architecture

Monolithic architecture is a unified model where all application components are tightly integrated into a single codebase.

Working Principle:

- All functionalities are executed within one application
- Components communicate through internal function calls
- Shared database is used

Advantages:

- Easy to develop and deploy
- Faster communication within system
- Lower initial cost

Disadvantages:

- Limited scalability
 - Difficult to update individual components
 - Entire system failure on single fault
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3.2 Microservices Architecture

Microservices architecture divides the application into smaller independent services.

Working Principle:

- Each service performs a specific function
- Services communicate via APIs
- Independent databases may be used

Advantages:

- High scalability
- Independent deployment
- Better fault isolation
- Flexibility in technology

Disadvantages:

- Complex system management
- Increased communication overhead
- Requires advanced DevOps practices

4. IMPLEMENTATION

4.1 Workflow Execution

The system follows these steps:

1. User sends request
 2. Request enters system through API gateway
 3. Authentication is performed
 4. Business logic processes request
 5. Database operations executed
 6. Response returned to user
-

4.2 Deployment Environment

- Cloud-based environment used
 - Docker for containerization
 - Kubernetes for orchestration
 - Load testing tools for simulation
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4.3 Monitoring and Logging

- Performance metrics tracked
 - CPU and memory usage monitored
 - Logs collected for debugging
 - Real-time monitoring ensures system stability
-

5. RESULTS AND ANALYSIS

5.1 Performance Analysis

- Monolithic:
 - Faster response (~120 ms) at low load
 - Performance drops at high load
 - Microservices:
 - Slightly higher latency (~180 ms)
 - Stable performance under heavy load (~260 ms)
-

5.2 Throughput Analysis

- Monolithic → ~850 requests/sec

- Microservices → ~1200 requests/sec

Microservices show better concurrency handling.

5.3 Resource Utilization

- Monolithic:
 - Lower CPU usage
 - Lower memory consumption
 - Microservices:
 - Higher CPU usage
 - Higher memory usage due to multiple services
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5.4 Scalability Analysis

- Monolithic:
 - Vertical scaling
 - Limited expansion
 - Microservices:
 - Horizontal scaling
 - Easily handles large workloads
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5.5 Fault Tolerance

- Monolithic:
 - Single failure affects entire system
- Microservices:

- Failure isolated to one service
 - System continues to function
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5.6 Setup Time

- Monolithic:
 - Simple and quick setup
 - Microservices:
 - Requires multiple configurations
 - More setup time
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6. DISCUSSION

The comparison highlights the trade-offs between simplicity and scalability.

- Monolithic architecture is suitable for:
 - Small-scale applications
 - Limited user base
- Microservices architecture is suitable for:
 - Large-scale systems
 - High traffic environments

Microservices provide better flexibility, but at the cost of increased complexity. Organizations must consider their requirements before choosing an architecture.

7. LIMITATIONS OF STUDY

- Simulated workloads may not fully represent real-world complexity

- Limited hardware and cloud configurations used
 - Communication delays in microservices may vary in different environments
 - Cost analysis not deeply explored
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8. CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

- Monolithic architecture is simple and efficient for small applications
 - Microservices architecture is highly scalable and fault tolerant
 - Microservices outperform monolithic systems in large-scale environments
 - Choice of architecture depends on system requirements
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8.2 Future Scope

- Development of hybrid architectures
- Integration of AI for system optimization
- Improved monitoring and debugging tools
- Advanced security mechanisms for distributed systems
- Cost optimization strategies for cloud deployment

